# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 2 | $-4x + 2y = 6$ becomes $-4x + 2y - 6 = 0$. |
| 2 | **(a)** Yes | $5(5) + (7)(3) = 46$, equal to 46. |
| 3 | **(b)** 5 | $6y = 34 - (2)(2)$ gives $y = 5$. |
| 4 | **(d)** a straight line | Fact. |
| 5 | **(c)** 4 | Compare with $y = mx + c$. |
| 6 | **(c)** parallel (or the same line) | Equal slopes. |
| 7 | **(a)** 4 | $1(2) + (7)k = 30$ gives $k = 4$. |
| 8 | **(b)** $\frac{14}{3}$ | Put x=0: $y = \frac{14}{3}$. |
| 9 | **(b)** 4 | $4x - 5y = 1$ becomes $4x - 5y - 1 = 0$. |
| 10 | **(d)** No | $6(7) + (5)(4) = 62$, not equal to 40. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $4x + 8y - 5 = 0$ (1); $a = 4$, $b = 8$, $c = -5$ (1). | 1+1 |
| 12 | Left side $= 7(4) + (2)(2) = 32$; Right side $= 32$ (1); Yes, it is a solution (1). | 1+1 |
| 13 | Substitute $x = 4$: $4(4) + (7)k = 44$; $7k = 28$ (1); $k = 4$ (1). | 1+1 |
| 14 | $-9x + 9y - 6 = 0$ (1); $a = -9$, $b = 9$, $c = -6$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $x = 0$: $6y = 36$ gives $y = 6$, so $(0, 6)$ (1); $y = 0$: $4x = 36$ gives $x = 9$, so $(9, 0)$ (2). | 1+2 |
| 16 | Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - (4)}{0 - (1)}$ (1); $= 2$ (2). | 1+2 |
| 17 | Left side $= 3(6) + (2)(2) = 22$ (1); Right side $= 22$ (1); Yes, it is a solution (1). | 1+1+1 |
| | **OR:** $x = 0$: $5y = 15$ gives $y = 3$, so $(0, 3)$ (1); $y = 0$: $5x = 15$ gives $x = 3$, so $(3, 0)$ (2). | 1+2 |
| 18 | Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - (1)}{4 - (0)}$ (1); $= \frac{3}{4}$ (2). | 1+2 |
| | **OR:** Left side $= 7(7) + (7)(7) = 98$ (1); Right side $= 98$ (1); Yes, it is a solution (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $6y = -7x + 43$ (1); $y = -\frac{7}{6}x + \frac{43}{6}$ (2); Slope $= -\frac{7}{6}$, $y$-intercept $= \frac{43}{6}$ (2). | 1+2+2 |
| | **OR:** $y = 11x + 31$ (1); For $x = 4$: $y = 11 \times 4 + 31$ (2); Fare ₹75 (2). | 1+2+2 |
| 20 | $6y = -x + 28$ (1); $y = -\frac{1}{6}x + \frac{14}{3}$ (2); Slope $= -\frac{1}{6}$, $y$-intercept $= \frac{14}{3}$ (2). | 1+2+2 |
| | **OR:** $y = 11x + 45$ (1); For $x = 9$: $y = 11 \times 9 + 45$ (2); Fare ₹144 (2). | 1+2+2 |

**Mistakes to avoid:**

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
- Saying a linear equation in two variables has one solution; it has infinitely many.
- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as $(y, x)$ instead of $(x, y)$.
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** $3$ | $6y = 42 - (6)(4)$ gives $y = 3$. |
| 2 | **(c)** infinitely many solutions | Each point on the line. |
| 3 | **(c)** $8$ | Compare with $y = mx + c$. |
| 4 | **(a)** a vertical line | All points have $x = 5$. |
| 5 | **(c)** $2$ | $3(1) + (1)k = 5$ gives $k = 2$. |
| 6 | **(c)** $\frac{10}{3}$ | Put y=0: $x = \frac{10}{3}$. |
| 7 | **(a)** $3$ | $3x - 6y = 6$ becomes $3x - 6y - 6 = 0$. |
| 8 | **(a)** Yes | $7(4) + (2)(1) = 30$, equal to $30$. |
| 9 | **(d)** $5$ | $4y = 45 - (5)(5)$ gives $y = 5$. |
| 10 | **(b)** the origin | $(0, 0)$ satisfies it. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Left side $= 5(1) + (7)(6) = 47$; Right side $= 38$ (1); No, it is not a solution (1). | 1+1 |
| 12 | Substitute $x = 3$: $7(3) + (6)k = 33$; $6k = 12$ (1); $k = 2$ (1). | 1+1 |
| 13 | $-2x - 3y - 8 = 0$ (1); $a = -2$, $b = -3$, $c = -8$ (1). | 1+1 |
| 14 | Left side $= 2(4) + (7)(1) = 15$; Right side $= 15$ (1); Yes, it is a solution (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $x = 0$: $4y = 60$ gives $y = 15$, so $(0, 15)$ (1); $y = 0$: $6x = 60$ gives $x = 10$, so $(10, 0)$ (2). | 1+2 |
| 16 | Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - (4)}{6 - (0)}$ (1); $= \frac{1}{3}$ (2). | 1+2 |
| 17 | Left side $= 3(6) + (4)(6) = 42$ (1); Right side $= 42$ (1); Yes, it is a solution (1). | 1+1+1 |
| | **OR:** $x = 0$: $2y = 28$ gives $y = 14$, so $(0, 14)$ (1); $y = 0$: $4x = 28$ gives $x = 7$, so $(7, 0)$ (2). | 1+2 |
| 18 | Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 - (2)}{5 - (2)}$ (1); $= -\frac{2}{3}$ (2). | 1+2 |
| | **OR:** Left side $= 1(4) + (3)(6) = 22$ (1); Right side $= 26$ (1); No, it is not a solution (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $5y = -7x + 33$ (1); $y = -\frac{7}{5}x + \frac{33}{5}$ (2); Slope $= -\frac{7}{5}$, $y$-intercept $= \frac{33}{5}$ (2). | 1+2+2 |
| | **OR:** $y = 10x + 55$ (1); For $x = 6$: $y = 10 \times 6 + 55$ (2); Fare ₹115 (2). | 1+2+2 |
| 20 | $7y = -3x + 19$ (1); $y = -\frac{3}{7}x + \frac{19}{7}$ (2); Slope $= -\frac{3}{7}$, $y$-intercept $= \frac{19}{7}$ (2). | 1+2+2 |
| | **OR:** $y = 19x + 27$ (1); For $x = 8$: $y = 19 \times 8 + 27$ (2); Fare ₹179 (2). | 1+2+2 |

**Mistakes to avoid:**

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
- Saying a linear equation in two variables has one solution; it has infinitely many.
- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as $(y, x)$ instead of $(x, y)$.
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** $2$ | Compare with $y = mx + c$. |
| 2 | **(c)** a horizontal line | All points have $y = -3$. |
| 3 | **(b)** $3$ | $1(1) + (2)k = 7$ gives $k = 3$. |
| 4 | **(b)** $7$ | Put x=0: $y = 7$. |
| 5 | **(d)** $2$ | $2x + 2y = 2$ becomes $2x + 2y - 2 = 0$. |
| 6 | **(d)** Yes | $2(5) + (-1)(3) = 7$, equal to 7. |
| 7 | **(d)** $4$ | $6y = 44 - (5)(4)$ gives $y = 4$. |
| 8 | **(a)** its steepness | Rise over run. |
| 9 | **(b)** $-1$ | Compare with $y = mx + c$. |
| 10 | **(d)** its coordinates satisfy the equation of the line | Definition. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Substitute $x = 4$: $5(4) + (4)k = 40$; $4k = 20$ (1); $k = 5$ (1). | 1+1 |
| 12 | $-9x + 4y - 5 = 0$ (1); $a = -9$, $b = 4$, $c = -5$ (1). | 1+1 |
| 13 | Left side $= 7(5) + (6)(4) = 59$; Right side $= 59$ (1); Yes, it is a solution (1). | 1+1 |
| 14 | Substitute $x = 5$: $3(5) + (2)k = 19$; $2k = 4$ (1); $k = 2$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $x = 0$: $5y = 35$ gives $y = 7$, so $(0, 7)$ (1); $y = 0$: $5x = 35$ gives $x = 7$, so $(7, 0)$ (2). | 1+2 |
| 16 | Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{6 - (0)}{4 - (3)}$ (1); $= 6$ (2). | 1+2 |
| 17 | Left side $= 2(5) + (7)(5) = 45$ (1); Right side $= 33$ (1); No, it is not a solution (1). | 1+1+1 |
| | **OR:** $x = 0$: $4y = 44$ gives $y = 11$, so $(0, 11)$ (1); $y = 0$: $4x = 44$ gives $x = 11$, so $(11, 0)$ (2). | 1+2 |
| 18 | Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{4 - (2)}{2 - (4)}$ (1); $= -1$ (2). | 1+2 |
| | **OR:** Left side $= 2(7) + (5)(5) = 39$ (1); Right side $= 27$ (1); No, it is not a solution (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $7y = -5x + 48$ (1); $y = -\frac{5}{7}x + \frac{48}{7}$ (2); Slope $= -\frac{5}{7}$, $y$-intercept $= \frac{48}{7}$ (2). | 1+2+2 |
| | **OR:** $y = 16x + 56$ (1); For $x = 5$: $y = 16 \times 5 + 56$ (2); Fare ₹136 (2). | 1+2+2 |
| 20 | $6y = -2x + 28$ (1); $y = -\frac{1}{3}x + \frac{14}{3}$ (2); Slope $= -\frac{1}{3}$, $y$-intercept $= \frac{14}{3}$ (2). | 1+2+2 |
| | **OR:** $y = 13x + 49$ (1); For $x = 9$: $y = 13 \times 9 + 49$ (2); Fare ₹166 (2). | 1+2+2 |

**Mistakes to avoid:**

- Finding the x-intercept by putting $x = 0$ (that is the y-intercept); for the x-intercept put $y = 0$.
- Writing the slope of $ax + by = c$ as $\frac{a}{b}$ instead of $-\frac{a}{b}$.
- Saying a linear equation in two variables has one solution; it has infinitely many.
- Treating $x = 5$ as a point; in the plane it is a vertical line.
- Plotting the pair as $(y, x)$ instead of $(x, y)$.
- Making a sign slip while transposing, e.g. $2x + y = 8 \Rightarrow y = 2x + 8$ instead of $y = -2x + 8$.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.